DeIR-Dual: Dual-Query Instruction Rewriting for Training-Free Instruction-Following Retrieval

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Abstract

Instruction-following retrieval requires systems to adapt their ranking when query instructions change. Existing query rewriting methods (e.g., HyDE, Query2Doc, DeepRetrieval) improve raw retrieval quality through semantic enhancement but lose instruction sensitivity (p-MRR ≈ 0). We propose DeIR-Dual, which decomposes an instruction into a positive enhancement query (Q_{pos}) and a negative exclusion query (Q_{minus}), then applies a training-free decision-layer scoring mechanism that estimates background leakage in the negative channel and penalizes only residual exclusion evidence. Our experiments show that (1) inference-time score calibration derives query-specific α_q and β_q from candidate-set score distributions, avoiding labeled validation data and test-set grid search; (2) DeIR-Dual achieves the strongest average instruction sensitivity and retrieval quality among the compared FollowIR methods; and (3) additional text-retrieval evaluations test whether the decision-layer formulation transfers beyond FollowIR.

Code:

1 Introduction

Information retrieval increasingly needs to answer more than a topical query. Users may ask for documents that satisfy a particular condition, avoid a confounding population, emphasize a desired aspect, or conform to a specified relevance criterion. In these settings, *instruction-following retrieval* (IF-IR) must rank documents by both their topical relation to the query and their consistency with the user's instruction. This is a stricter requirement than conventional semantic matching: two documents can discuss the same topic, yet only one can satisfy the condition that makes it useful to the user [7]. More broadly, instruction tuning has made language models responsive to natural-language directives [13, 23, 17]; IF-IR requires that same control to determine document relevance rather than generated text.

Recent benchmarks make this distinction measurable. FollowIR [24] pairs a query with detailed relevance instructions and evaluates whether a retriever changes its ranking when the instruction changes. Its results expose a persistent failure mode: models that are strong at ordinary retrieval often continue to favor documents with the right topic or keywords while underweighting the fine-grained criteria that determine relevance. The problem is not limited to a particular linguistic form. Positive requirements, contextual qualifications, and explicit exclusions can all change which documents should be preferred within the same topical candidate set.

Existing methods either adapt the retriever with instruction supervision or intervene at inference time. Promptriever [25], InF-IR [29], and Dual-View Training [27] adapt instruction-sensitive retrievers using additional training data and optimization. Training-free query rewriting, such as HyDE [4], Query2Doc [22], DeepRetrieval [9], and RAG-Fusion [16], instead retrieves with enhanced query representations. It improves semantic coverage, but supplies no document-level rule for resolving competing instruction signals.

Explicit exclusion illustrates why this decision is difficult. A document that violates an exclusion is often not irrelevant: it can share the query's topic, entities, and vocabulary with a desirable document. Penalizing it solely for matching an exclusion description is also unreliable, because the same topical background can induce high negative-channel similarity. A more structured inference-time approach makes this explicit at the representation layer: DEO [8] decomposes text queries and optimizes their embeddings, while Negation Matters [15] decomposes visual-language queries and applies embedding-space repulsion with context anchoring. These methods show that frozen backbones can be made negation-aware, but resolve competing signals by altering query directions or similarities in the embedding space. We instead ask whether a frozen retriever's candidate set already provides sufficient local evidence to make the decision at the document level.

We propose DeIR-Dual, a training-free decision-layer reranker for IF-IR. It uses instruction decomposition to construct three auxiliary scoring views, rather than replacing the original query with a single rewritten representation: a base query (Q_{base}) that preserves the full query and instruction, a positive refinement query (Q_{pos}) that exposes affirmative evidence, and a negative exclusion query (Q_{minus}) that exposes evidence for suppression. The views are not a taxonomy of instruction semantics; together, they let the reranker retain context while separately assessing helpful and potentially conflicting signals. Crucially, DeIR-Dual estimates the amount of negative-channel score expected from background relevance, penalizes only residual exclusion evidence above a query-adaptive boundary, and uses that same boundary to gate the positive reward. Label-free candidate-set calibration derives the reward and penalty scales at inference time, without retriever updates, relevance labels, or benchmark-specific coefficient search.

We evaluate the resulting decision-layer formulation on complementary settings. FollowIR tests whether rankings respond to changed instructions while preserving retrieval quality, whereas NegConstraint [26] tests transfer to retrieval queries with explicit logical exclusions. Across the compared FollowIR methods, DeIR-Dual achieves the strongest average instruction sensitivity and retrieval quality, and the same formulation is evaluated on NegConstraint without dataset-specific retraining. A remaining limitation is **semantic entanglement**—strong overlap between the base and exclusion channels— which can make residual exclusion evidence difficult to isolate. In summary, our contributions are: **(i)** a query-adaptive residual exclusion boundary and safety-gated score composition for document-level instruction resolution; **(ii)** a label-free query-local calibration scheme for adapting this decision layer to candidate-set score scales; and **(iii)** an empirical evaluation spanning changed-instruction retrieval and explicit negative-constraint retrieval.

2 Related Work

2.1 Supervised Instruction-Following Retriever Adaptation

Instruction-following retriever adaptation is a primary route to IF-IR. Instruction-tuned encoders such as INSTRUCTOR [18] and TART [1] train embedding models with task descriptions. More directly, Promptriever [25] constructs instance-level instruction data and fine-tunes a language-model retriever to be promptable at retrieval time; InF-IR [29] likewise uses instruction supervision to train a retriever. Dual-View Training [27] is a training-based extension that synthesizes polarity-reversed views of a document pair. FollowIR [24] formalizes the resulting requirement through original/changed instruction pairs and p-MRR. Together, these works establish that instruction sensitivity can be learned, but rely on additional data, optimization, and retriever updates.

Related LLM embedding approaches, including E5-Mistral and GritLM, use instructions to condition general representation learning and are typically assessed on broad embedding and retrieval suites such as MTEB and BEIR [20, 11, 12, 19]. Their instructions usually identify a task or format, whereas IF-IR requires the relevance definition itself to change from one query instance to the next.

2.2 Training-Free Instruction Resolution

LLM-based query rewriting improves retrieval by making the user's information need more explicit before matching. HyDE [4] retrieves with hypothetical documents, Query2Doc [22] enriches queries with pseudo-documents, and RAG-Fusion [16] combines multiple generated query variants. More recent rewriting systems learn the rewriting policy or optimize retrieval-oriented generation, including RAG-QR [10] and DeepRetrieval [9]. These methods primarily improve recall or semantic coverage by retrieving with enhanced query representations. Even when multiple rewrites are generated, the variants usually remain positive descriptions of relevance rather than independent document-level evidence for what should be suppressed.

Exclusion constraints are a particularly challenging form of instruction following. In exclusion queries, the hardest negatives are not irrelevant documents but topically relevant documents that violate the constraint. Such documents may match the affirmative intent and share substantial background with the query, yet violate the user's "not", "without", or "exclude" condition. This makes positive query enhancement alone less suited to the problem: it can strengthen topical relevance without identifying which relevant-looking documents should be suppressed.

Training-free decomposition methods make the missing exclusion signal explicit, but differ in where they resolve its conflict with positive relevance. DEO [8] decomposes text queries into positive and negative components and optimizes the query embedding. In vision-language retrieval, Negation Matters [15] similarly decomposes affirmative and negated content, then applies embedding-space repulsion and context anchoring over a frozen CLIP backbone. These methods demonstrate that training-free negation handling is possible, but their corrections are expressed as modified query directions or representation-space similarities. DeIR-Dual instead keeps the retriever and its embeddings fixed, resolving competing context, positive, and negative evidence at the candidate-set decision layer through background-leakage estimation and query-local calibration.

2.3 Constraint-Aware Retrieval

One line of work addresses negative constraints and complex logical queries by making constraint structure explicit. Neural-symbolic retrieval methods such as NS-IR [26] convert natural-language queries and documents into first-order logic forms and enforce logical consistency through reranking objectives such as logic alignment and connective constraints. OrLog [6] instead decomposes complex queries into atomic predicates, estimates predicate-level plausibility with LLMs, and applies probabilistic reasoning to infer query satisfaction. These routes offer interpretable ways to represent logical constraints, but they introduce a different dependency: natural-language retrieval must be converted into logical or predicate-level representations, or coupled with symbolic/probabilistic reasoning objectives. Our method stays in the dense score space and does not require a logical parser, symbolic inference module, or probabilistic reasoning engine.

2.4 Query-Local Score Calibration and Test-Time Adaptation

The calibration component of DeIR-Dual is related to post-hoc calibration and local similarity correction. Platt scaling and temperature scaling transform model outputs after training while keeping model parameters fixed [14, 5], but are typically estimated from labeled validation data for confidence calibration. Local scaling methods such as CSLS [2] show a complementary principle: raw similarities can contain local bias, and neighborhood statistics can improve comparability across candidates. Our setting differs from both lines because it requires label-free calibration of ranking scores for each query.

Information retrieval also provides a direct precedent for using unlabeled query-local evidence. Query performance prediction estimates the expected reliability of a ranking without relevance judgments, and recent studies show that neural rankers exhibit query-specific behaviors that are difficult to capture with traditional predictors [3]. Pseudo-relevance feedback similarly uses the top-ranked set as local evidence, including in dense retrieval settings where feedback representations are extracted from first-pass results [21]. DeIR-Dual adopts the same broad test-time principle but uses the candidate set differently: it does not predict query difficulty or expand the query, but estimates the score scales of the base, positive, and negative channels.

Recent inference-only calibration work further shows that unlabeled test batches can be used to estimate and correct score biases without supervised validation labels [28]. In contrast, DeIR-Dual does not tune prompts, update model parameters, or learn global calibration constants. It keeps the retriever fixed and uses candidate-set statistics only to calibrate decision-layer reward and penalty scales.

3 Method: DeIR-Dual

3.1 Problem Formulation

We consider two-stage inference with a frozen dense retriever. Given a query q and instruction I, the first stage retrieves a top-K candidate set $\mathcal{C}_q$ with the full-context view Q_{base}. DeIR-Dual then reorders only $\mathcal{C}_q$ according to topical relevance and the constraints expressed in I, without retriever updates or labeled calibration data. This separation preserves first-stage recall while concentrating constraint resolution on the documents that compete at the top of the ranking. The difficulty is not simply to make the instruction more explicit: a document that violates an exclusion can still be highly relevant to the query topic. Its negative-channel score therefore mixes constraint-violation evidence with similarity caused by shared background.

DeIR-Dual resolves this ambiguity at the decision layer. It retains a frozen retriever and separates the available evidence into contextual, affirmative, and exclusion views. It then estimates how much negative-channel similarity is explained by background relevance and uses the residual only for candidate-level decisions.

3.2 Triple-Query Scoring Views

An LLM decomposer constructs three short retrieval-style queries from (q, I): Q_{base}, containing the original query and full instruction; Q_{pos}, retaining affirmative constraints; and Q_{minus}, containing only excluded content. In particular, Q_{minus} names the excluded content rather than retaining a negation operator (e.g., "without X" becomes "X"). These are operational scoring views, not a taxonomy of instruction semantics. The base view retains context that can be lost in an affirmative/exclusion split, while the other two views isolate evidence used by the decision layer. If no exclusion is present, the decomposer returns $Q_{\text{minus}} = \text{[NONE]}$ and the negative branch is skipped.

Table 1 illustrates the decomposition. The exclusion view is deliberately a retrievable description of the out-of-scope content: high similarity to it is evidence to be assessed against topical background in the next stage, rather than an instruction executed directly in the embedding space.

Let f_q and f_d denote the query and document encoders of the unchanged retriever. For each $d \in \mathcal{C}_q$, we compute

$$\begin{aligned} S_{\text{base}}(d) &= \cos(f_q(Q_{\text{base}}), f_d(d)), \qquad S_{\text{pos}}(d) = \cos(f_q(Q_{\text{pos}}), f_d(d)), \\ S_{\text{neg}}(d) &= \cos(f_q(Q_{\text{minus}}), f_d(d)). \end{aligned} \tag{1}$$

Document embeddings are computed once and reused for all views.

Table 1: Illustrative operational decomposition. Q_{base} preserves the complete request; Q_{pos} states what should qualify as relevant; and Q_{minus} describes only the content to exclude.

View	Text
Input	*What is the ongoing status of the Three Gorges Project?* Relevant documents report its completion date, total cost, or electrical output; social, political, and ecological impacts are out of scope.
Q_{base}	What is the ongoing status of the Three Gorges Project? Relevant documents report its completion date, total cost, or electrical output; social, political, and ecological impacts are out of scope.
Q_{pos}	Status of the Three Gorges Project: completion date, total cost, and electrical output.
Q_{minus}	Social, political, and ecological impacts of the Three Gorges Project.

3.3 Query-Adaptive Exclusion Boundary

The base and positive channels provide topical relevance and instructional refinement. The negative channel is more delicate: a high $S_{\text{neg}}(d)$ can indicate a true constraint violation, but it can also arise because relevant and excluded documents share entities, topic, or background context. DeIR-Dual therefore treats exclusion as a **residual boundary estimation** problem. Negative-channel similarity is decomposed into background leakage and residual exclusion evidence. We estimate the background component from the current query's candidate distribution, subtract it from the negative-channel score, and penalize residual evidence only as it approaches and exceeds a robust query-specific boundary.

Background Leakage Estimation

Our treatment follows the same broad intuition as local scaling methods for nearest-neighbor retrieval: raw similarity can contain a query-local bias that should be removed before comparison [2]. In our setting, this bias is background-induced similarity in the negative channel. We use a simple query-local linear leakage model to estimate the part of the negative-channel score that is expected from background relevance. Let μ_b, σ_b be the mean and standard deviation of S_{base} over the candidate set, and let μ_n, σ_n be the corresponding statistics for S_{neg}. We first standardize the base score,

$$z_b(d) = \frac{S_{\text{base}}(d) - \mu_b}{\sigma_b}, \tag{2}$$

and measure the query-level overlap between the base and exclusion channels as

$$c_q = \cos(f_q(Q_{\text{base}}), f_q(Q_{\text{minus}})). \tag{3}$$

We use a query-local affine baseline for the negative score:

$$\widehat{S}^{\text{bg}}_{\text{neg}}(d) = \mu_n + \sigma_n \, c_q \, z_b(d). \tag{4}$$

This baseline transfers base-score variation into the negative-channel scale according to base/exclusion semantic overlap. It is a query-local control for score co-variation, rather than a relevance model in its own right.

Residual Exclusion Boundary

We define the residual negative score as the excess negative-channel evidence beyond the background estimate:

$$R_{\text{neg}}(d) = S_{\text{neg}}(d) - \widehat{S}^{\text{bg}}_{\text{neg}}(d). \tag{5}$$

Positive residuals indicate that the document matches the exclusion query more strongly than expected from background relevance alone. We define a robust query-adaptive residual margin:

$$a_q = \text{MAD}(\mathcal{R}_q), \qquad m_q = \lambda a_q, \quad \mathcal{R}_q = \{R_{\text{neg}}(d) : d \in \mathcal{C}_q\}, \tag{6}$$

where $\mathcal{C}_q$ denotes the retrieved candidate set. Residuals above m_q provide stronger evidence for exclusion. We use Softplus as a smooth approximation to a hinge penalty:

$$\text{penalty}(d) = \alpha_q \ \text{Softplus}(R_{\text{neg}}(d) - m_q)\,. \tag{7}$$

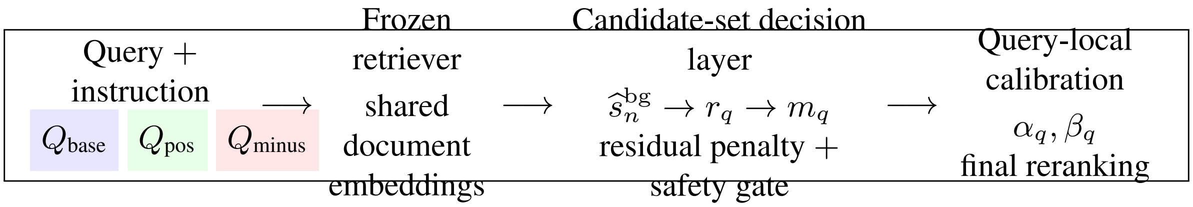

Figure 1: DeIR-Dual uses Q_{base} for first-stage retrieval and keeps the retriever frozen. Three scoring views provide contextual, affirmative, and exclusion evidence for the same top-K candidate set. The decision layer removes background-induced negative similarity, separately forms a residual penalty and a safety gate, and calibrates their scales from unlabeled candidate-set scores.

The penalty grows smoothly as residual exclusion evidence approaches and exceeds the boundary, while the residual construction prevents the negative channel from penalizing documents merely because they share topical background with the exclusion query. The scale α_q is derived by score calibration in §3.4.

Safety-Gated Positive Reward

The positive channel can improve recall for documents satisfying the instruction, but a raw positive reward can also amplify documents that violate the exclusion constraint. We therefore gate the positive reward by residual evidence relative to the background expectation:

$$g_q(d) = 1 - \sigma\left(\kappa \frac{R_{\text{neg}}(d)}{a_q}\right). \tag{8}$$

The penalty margin and the gate have distinct roles: m_q determines when an extra exclusion penalty becomes active, whereas the gate begins to reduce positive reward once the residual exceeds the background expectation. The fixed shape parameter κ controls the sharpness of this transition.

Final score. The final decision-layer score combines base relevance, safety-gated positive refinement, and residual exclusion penalty:

$$S_{\text{final}}(d) = S_{\text{base}}(d) + \beta_q\, S_{\text{pos}}(d)\, g_q(d) - \alpha_q\ \text{Softplus}(R_{\text{neg}}(d) - m_q)\,. \tag{9}$$

This nonlinear score composition implements a query-adaptive exclusion boundary at ranking time without modifying the dense retriever. The parameters α_q and β_q are derived from the score distributions of the current query, while λ and κ are fixed shape parameters controlling boundary offset and gate sharpness. When an instruction has no exclusion, we omit the negative branch and rank by $S_{\text{base}}(d) + \beta_q S_{\text{pos}}(d)$. Numerical safeguards and missing-view handling are deferred to Appendix F.

3.4 Inference-Time Score Calibration

The scoring formula in Eq. 9 contains two query-level calibration factors: α_q for the residual exclusion penalty and β_q for the positive refinement reward. Rather than grid-searching these values on labeled queries, we derive them at reranking time from the score distributions of the current candidate set. This converts parameter selection into query-conditioned score calibration: the correction terms are placed on a magnitude comparable to the base retriever without changing the retriever or using relevance labels. The goal is not to fit a benchmark-specific optimum, but to make the exclusion boundary and positive reward portable across retrievers with different score scales.

Candidate-Set Score Calibration

The base, positive, and negative channels are produced by the same dense retriever but have different score distributions because they encode different query intents. We therefore treat α_q and β_q as post-hoc scale factors for decision-layer reranking. Unlike validation-set calibration methods such as Platt scaling and temperature scaling [14, 5], our calibration is label-free, query-local, and operates on ranking scores. The only inputs are the channel scores over the current candidate set $\mathcal{C}_q$.

Let the residual excess over the boundary be

$$o_q(d) = R_{\text{neg}}(d) - m_q, \tag{10}$$

and define the residual-exceeding subset as

$$\mathcal{A}_q = \{d \in \mathcal{C}_q : o_q(d) > 0\}. \tag{11}$$

These are the documents whose negative residual exceeds the robust boundary and therefore carries residual exclusion evidence. We write the corresponding penalty magnitude as

$$P_q(d) = \mathrm{Softplus}(o_q(d)). \tag{12}$$

We derive α_q and β_q by active-region scale matching. The penalty is matched to base scores on the penalty-active set:

$$\alpha_q = \frac{\mathrm{Avg}_{d \in \mathcal{A}_q} S_{\text{base}}(d)}{\mathrm{Avg}_{d \in \mathcal{A}_q} P_q(d)}. \tag{13}$$

For the positive channel, we use the complementary safe subset

$$\mathcal{B}_q = \{d \in \mathcal{C}_q : o_q(d) \leq 0\}. \tag{14}$$

The reward scale is computed from the raw positive channel on the safe set. Keeping $g_q(d)$ out of this calibration prevents scale alignment from compensating for the safety gate:

$$\beta_q = \frac{\max_{d \in \mathcal{B}_q} S_{\text{base}}(d)}{\mathrm{Avg}_{d \in \mathcal{B}_q} S_{\text{pos}}(d)}. \tag{15}$$

Numerical safeguards for empty active sets and near-zero denominators are fixed before evaluation and described in Appendix F. Crucially, α_q and β_q use only the unlabeled score distribution of the current candidate set; no relevance labels, validation queries, or benchmark-level coefficient search enter Eqs. 13–15. In contrast, λ and κ are fixed shape constants for the residual margin and gate sharpness, respectively; their values and sensitivity are reported in Appendix F.

4 Experiments

4.1 Experimental Setup

We evaluate DeIR-Dual as an inference-time reranker over the top-K candidate set produced by a frozen base retriever. Every comparison uses the same candidate set for its base model; the three query views share the retriever's query template, and document embeddings are computed once. The decomposer, prompting configuration, candidate depth, and fixed shape constants λ, κ are held fixed within each evaluation protocol; only α_q, β_q are derived from the current unlabeled candidate set. Full configuration details are in Appendix F.

FollowIR [24] is our primary benchmark for changed-instruction retrieval on Robust04, News21, and Core17. We report its instruction-sensitivity p-MRR together with MAP/nDCG and the benchmark's aggregate Score. NegConstraint [26] provides a complementary transfer test in which exclusion is expressed directly as a retrieval constraint; there we report MAP and nDCG@10. Unless otherwise noted, methods are used without retriever fine-tuning, relevance labels, or benchmark-level coefficient search.

4.2 Main Results

Instruction Following on FollowIR

Table 2 reports representative results on FollowIR. It separates the frozen base retriever, a supervised instruction-adaptation reference (Promptriever), and representative inference-time methods. The direct comparison is therefore among the inference-time methods, whereas Promptriever indicates the performance of the data-construction-and-fine-tuning route. The complete comparison is deferred to Appendix A.

Portability Across Frozen Retrievers

Table 3 evaluates DeIR-Dual across frozen retrievers with different score scales and embedding spaces. For each retriever, reranking is performed on the same top-K candidate set returned by the base model. The same shape constants are used for all retrievers, while α_q and β_q are derived from each query's candidate scores. This setting tests whether query-local calibration is tied to a single encoder or can be applied as a portable decision-layer reranker over different frozen retrievers.

Method	Core17		Robust04		News21		Average	
	MAP	p-MRR	MAP	p-MRR	nDCG	p-MRR	Score	p-MRR
RepLLaMA (base)	20.6	+1.3	24.0	-8.9	24.5	-1.8	23.0	-3.1
Promptriever†	21.6	+15.4	28.3	+11.7	28.5	+6.4	26.1	+11.2
HyDE	23.6	+6.5	24.9	-2.9	21.4	+0.7	23.4	+1.4
Query2Doc	25.9	+8.0	27.9	-7.5	24.9	-3.8	26.2	-1.1
INF-X Full	27.0	+7.2	32.1	-1.4	22.3	+4.4	27.0	+3.4
DeIR-Dual	**26.0**	**+17.6**	**28.6**	**+8.3**	**32.3**	**+18.7**	**29.0**	**+14.9**

Table 2: Representative FollowIR comparison. Promptriever† is a supervised instruction-trained reference; the other non-base baselines are inference-time methods. All metrics are percentages ($\times 100$); the complete baseline table is in the appendix.

Retriever	Base Retriever		DeIR-Dual	
	Score	**p-MRR**	**Score**	**p-MRR**
E5-Mistral	38.4	−2.8	26.3	16.0
BGE	31.2	−2.2	21.1	19.5
GritLM	4.0	1.4	2.4	1.4
RepLLaMA	37.4	−3.3	25.2	20.3

Table 3: Cross-retriever results on FollowIR. Every row uses the same decomposer, candidate depth, and fixed λ, κ; only α_q, β_q are derived from the current candidate set.

Method	**MAP**	**nDCG@10**
Base retriever	70.8	77.7
+ Positive channel	76.2	81.6
+ Raw negative penalty	78.5	83.5
DEO (BGE-M3)†	73.8	79.5
DeIR-Dual	75.0	80.4

Table 4: Transfer to NegConstraint [26]. DEO† is the reported BGE-M3 result from its arXiv preprint [8]; it is an embedding-space reference rather than a matched reranking comparison. All scores are percentages ($\times 100$).

Transfer to Negation-Sensitive Benchmarks

Table 4 evaluates transfer to NegConstraint, a text-retrieval benchmark where exclusion is expressed directly as a retrieval constraint. Following the NegConstraint protocol [26], we report the total MAP and nDCG@10 in the main text and defer formulation-level results to Appendix D. We also include DEO [8], a close training-free negation-aware retrieval baseline. The method is transferred without dataset-specific retraining or coefficient search: the same scoring mechanism and fixed shape constants are used, while α_q and β_q are derived from unlabeled candidate-set scores. This setting tests whether the residual boundary mechanism remains useful beyond FollowIR's changed-instruction protocol. The formulation-level breakdown is reported in Appendix D.

4.3 Ablation Study

Table 5 tests two competing explanations for the main result. The first block asks whether positive rewriting or a raw negative penalty is sufficient; the second asks whether the score scales can be fixed or calibrated globally. We report the aggregate FollowIR metrics here and defer per-collection breakdowns and secondary variants to Appendix B.

We fix λ and κ across all retrievers and datasets. Sensitivity, candidate-depth, efficiency, and secondary-design analyses are reported in Appendix B.

Variant	Avg. Score	Avg. p-MRR
Exclusion mechanism		
Base retriever	23.81	-3.35
+ Positive channel	25.04	19.74
+ Raw negative penalty	26.61	1.14
+ Residual boundary	26.86	5.29
Full DeIR-Dual	25.33	**20.14**
Score calibration		
Fixed α, β	25.24	18.53
Global calibration	25.14	19.90
Query-local calibration	25.23	20.31
Grid-search oracle[‡]	24.21	**22.29**

Table 5: Compact FollowIR ablation. The first block contrasts competing exclusion mechanisms; the second changes only the selection of α, β. [‡]Oracle tuning is an analysis upper bound, not a deployable method. All metrics are percentages ($\times 100$).

	Core17		Robust04		News21		Average	
Method	MAP	p-MRR	MAP	p-MRR	nDCG	p-MRR	Score	p-MRR
DeepRetrieval	21.5	+6.6	25.9	-5.3	23.9	-2.2	23.8	-0.3
HyDE	23.6	+6.5	24.9	-2.9	21.4	+0.7	23.4	+1.4
Query2Doc	25.9	+8.0	27.9	-7.5	24.9	-3.8	26.2	-1.1
RAG-Fusion	21.9	+5.4	21.1	-8.1	21.0	+1.8	21.3	-0.3
RAG-QR	22.3	+2.3	27.6	-10.3	22.9	-2.1	24.3	-3.4
ConvSearch-R1	22.3	+3.0	25.9	-2.5	22.0	+0.2	23.4	+0.2
INF-X Full	27.0	+7.2	32.1	-1.4	22.3	+4.4	27.0	+3.4
BGE-Reasoner-Rewriter	21.5	+0.7	TBD	TBD	20.5	+3.4	TBD	TBD
ReFeed-8B	18.3	-6.6	23.3	-1.2	22.4	-0.4	21.2	-2.7
TongSearch-QR	21.7	+3.3	24.5	-4.6	22.6	+0.2	22.7	-0.4
DeIR-Dual	**26.0**	**+17.6**	**28.6**	**+8.3**	**32.3**	**+18.7**	**29.0**	**+14.9**

Table 6: Complete FollowIR comparison for inference-time baselines. All metrics are percentages ($\times 100$).

5 Conclusion and Limitations

This paper presented DeIR-Dual, a frozen-retriever decision-layer reranking method for text retrieval under exclusion-sensitive instructions. The central idea is to avoid treating negative-query similarity as uniformly harmful. DeIR-Dual first uses triple-query decomposition to separate contextual relevance, positive refinement, and exclusion evidence; it then estimates a query-adaptive residual exclusion boundary over the candidate set and applies label-free query-local score calibration to place residual penalties and gated positive rewards on comparable scales. The current FollowIR results show improvements in instruction sensitivity and aggregate retrieval quality over inference-time rewriting and reranking baselines. The cross-retriever, transfer, ablation, and appendix analyses characterize how these effects vary across frozen retrievers, datasets, mechanism variants, and fixed shape constants.

The scope of DeIR-Dual is bounded by several limitations. It depends on LLM-generated decompositions, so ambiguous or underspecified instructions can propagate errors into the positive and negative channels. High-overlap cases, where desired and excluded semantics are tightly entangled, remain difficult because residual exclusion evidence can be hard to distinguish from background relevance. The fixed shape constants λ and κ are shared across settings, so their portability is assessed by the appendix sensitivity analysis. Finally, because DeIR-Dual reranks an existing candidate set rather than retrieving new documents, its effectiveness depends on the base retriever placing relevant documents in $\mathcal{C}_q$; missed candidates cannot be recovered by the decision-layer scorer.

A Complete FollowIR Comparison

Table 6 expands Table 2 to all evaluated inference-time baselines. Values unavailable from a baseline report are retained as TBD rather than imputed.

Variant	Core17		Robust04		News21		Average	
	MAP	p-MRR	MAP	p-MRR	nDCG	p-MRR	Score	p-MRR
Base retriever	23.21	1.07	25.71	−9.27	22.52	−1.84	23.81	−3.35
+ Positive channel	24.82	27.44	23.25	9.57	27.04	22.21	25.04	19.74
+ Raw negative penalty	26.56	0.49	24.47	1.77	28.79	1.15	26.61	1.14
+ Residual boundary	26.88	4.74	23.87	5.11	29.82	6.03	26.86	5.29
Full DeIR-Dual	24.70	28.17	22.88	9.61	28.41	22.65	25.33	20.14

Table 7: Per-collection exclusion-mechanism ablation on FollowIR.

Scale selection	Core17		Robust04		News21		Average	
	MAP	p-MRR	MAP	p-MRR	nDCG	p-MRR	Score	p-MRR
Fixed α, β	24.67	27.26	23.59	6.96	27.46	21.38	25.24	18.53
Global calibration	24.70	27.92	23.31	9.24	27.40	22.55	25.14	19.90
Query-local calibration	24.96	28.15	22.85	10.02	27.87	22.75	25.23	20.31
Grid-search oracle	22.17	31.68	22.77	10.20	27.70	25.00	24.21	22.29

Table 8: Per-collection calibration ablation on FollowIR.

Variant	Change from Full DeIR-Dual	Avg Score	Avg p-MRR	Runtime
Full DeIR-Dual	None	25.3	20.1	<1 s
w/o safety gate	Remove $g_q(d)$ from positive reward	26.6	1.1	<1 s
α_q only	Query-local penalty scale, fixed reward scale	25.0	19.7	<1 s
β_q only	Query-local reward scale, fixed penalty scale	25.3	4.3	<1 s
Mean/std boundary	Replace median/MAD boundary with mean/std	25.2	19.3	<1 s
Raw s_n boundary	Estimate boundary on raw negative similarity	26.9	5.3	<1 s

Table 9: Diagnostic ablations for secondary design choices. The table keeps the same frozen retriever and candidate pool as Table 5.

Analysis	Setting	Avg Score	Avg p-MRR	Latency	Notes
λ sensitivity	$\lambda \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$	25.3 ± 0.0	16.7 ± 0.0	<1 s	Insensitive
κ sensitivity	$\kappa \in \{1, 2, 4, 8, 10, 15, 20\}$	25.3 ± 0.8	20.1 ± 3.5	<1 s	Trade-off
Candidate set size	$K \in \{10, 50, 100, 200\}$	TBD	TBD	TBD	To be measured
Standard retrieval control	BEIR/TREC subset	TBD	TBD	TBD	To be measured
Failure subset	High semantic-overlap queries	TBD	TBD	TBD	To be analyzed

Table 10: Sensitivity, efficiency, and robustness analyses. TBD entries are explicit placeholders for matched evaluation runs.

B Additional Ablations and Sensitivity

Tables 7 and 8 give the per-collection breakdowns for the compact ablation in the main text. Table 9 then separates secondary design choices that should not be conflated with the two central mechanisms.

Table 10 records robustness and efficiency analyses. TBD marks results retained for experiments that have not yet been run; no values are imputed.

C Standard Retrieval Preservation

This control asks whether the decision layer perturbs ordinary retrieval when no exclusion is present. We evaluate a representative BEIR subset covering biomedical, financial, argumentative, scientific, and duplicate- question retrieval. For each backbone, we compare the frozen base query, the same query paired with a task-neutral retrieval instruction, and DeIR-Dual with Q_{minus} = [NONE]. The latter condition uses the same candidate-generation protocol as the neutral-instruction condition and applies no exclusion penalty. We report average nDCG@10 and MAP@100 over the

Backbone	Base query		+ Neutral instruction		+ DeIR-Dual	
	nDCG@10	MAP@100	nDCG@10	MAP@100	nDCG@10	MAP@100
RepLLaMA	TBD	TBD	TBD	TBD	TBD	TBD
E5-Mistral	TBD	TBD	TBD	TBD	TBD	TBD
BGE	0.613	0.429	0.613	0.429	0.601	0.418

Table 11: Standard-retrieval preservation on a representative BEIR subset (TREC-COVID, NFCorpus, FiQA-2018, ArguAna, SciFact, and Quora). A neutral instruction isolates input-format effects; the final condition tests whether DeIR-Dual changes ordinary ranking when no exclusion is active. TBD entries are to be filled from matched runs.

Method	$A - a$		$(A - a) \cup B$		$(A - a) \cup (B - b)$		Total	
	MAP	nDCG@10	MAP	nDCG@10	MAP	nDCG@10	MAP	nDCG@10
BM25	32.1	34.6	31.2	34.7	29.3	31.5	31.4	33.7
Contriever	34.8	36.6	32.1	33.3	30.9	32.7	31.8	35.7
BGE	37.9	40.5	34.8	36.8	33.7	34.8	36.3	40.8
HyDE	50.7	55.3	48.6	51.5	45.7	50.6	47.8	53.1
InteR	52.6	55.8	50.3	48.7	51.5	49.3	52.3	54.5
NS-IR (GPT-4o)	54.7	57.9	53.3	54.2	51.7	53.7	53.3	56.5
BGE-S	TBD	TBD	TBD	TBD	TBD	TBD	67.0	73.7
BGE-S w/ DEO	TBD	TBD	TBD	TBD	TBD	TBD	73.0	78.0
BGE-L	TBD	TBD	TBD	TBD	TBD	TBD	63.0	71.4
BGE-L w/ DEO	TBD	TBD	TBD	TBD	TBD	TBD	73.3	78.8
BGE-M3	TBD	TBD	TBD	TBD	TBD	TBD	63.7	72.5
BGE-M3 w/ DEO	TBD	TBD	TBD	TBD	TBD	TBD	73.8	79.5
Base retriever	74.5	80.6	62.7	71.4	TBD	TBD	70.8	77.7
+ Positive channel	81.4	85.5	64.8	73.2	TBD	TBD	76.2	81.6
+ Raw negative penalty	83.7	87.4	67.3	75.0	TBD	TBD	78.5	83.5
DeIR-Dual	80.6	84.4	62.8	71.6	TBD	TBD	75.0	80.4

Table 12: Formulation-level NegConstraint results. ACL reference results are taken from the NegConstraint paper [26]. DEO results are total-only numbers reported in its arXiv preprint [8]. All scores are percentages ($\times$100).

selected BEIR tasks; these ranking metrics are nontrivial under candidate-set reranking, unlike candidate recall at an unchanged cutoff.

If the neutral-instruction and DeIR-Dual columns remain close to the base-query column, the experiment supports the claim that DeIR-Dual adds constraint resolution without systematically degrading generic retrieval. Any material degradation should instead be reported as a deployment boundary and analyzed by backbone or task category.

D Additional NegConstraint Results

Table 12 provides the formulation-level NegConstraint results corresponding to the total scores reported in Table 4. DEO reports only total scores in the referenced preprint, so unavailable formulation-level entries are marked TBD.

E Qualitative Analysis

TBD: Add one successful decomposition and one semantic-entanglement case. For each case, report the original query and instruction, the three generated views, representative candidate documents, their channel scores, and the resulting ranking change.

F Implementation Details for Decomposition and Calibration

The decomposer receives the original query and instruction and returns a JSON object with an affirmative field and an exclusion field. It is prompted to keep the affirmative field as a fluent retrieval query, to express exclusions as content rather than as logical negation tokens, and to return [NONE] when the instruction contains no exclusion. Q_{base} concatenates the original query and full instruction, while Q_{pos} and Q_{minus} are the two returned fields. Malformed outputs are retried with the same format constraint; a missing exclusion is treated as [NONE]. The exact prompt, model identifier, decoding configuration, and generated query files are included with the artifact.

All document representations are computed once with the base retriever's document template, and all three query views use its query template. For numerical stability, standard deviations and MAD values are lower-bounded by 10^{-6}. If the active or safe subset is empty, or a calibration denominator is below this tolerance, the corresponding scale retains its fixed fallback value. Derived α_q and β_q are clipped to $[0.05, 5.0]$. The shape constants λ and κ are fixed before evaluation, shared by all datasets and retrievers, and never selected using relevance labels or evaluation metrics. The released evaluation configuration records their single reported values and the candidate-set depth for each experiment.

References

[1] Akari Asai et al. Task-aware retrieval with instructions. In *arXiv preprint arXiv:2211.09260*, 2022.

[2] Alexis Conneau, Guillaume Lample, Marc'Aurelio Ranzato, Ludovic Denoyer, and Hervé Jégou. Word translation without parallel data. In *International Conference on Learning Representations*, 2018.

[3] Guglielmo Faggioli, Thibault Formal, Stefano Marchesin, Stéphane Clinchant, Nicola Ferro, and Benjamin Piwowarski. Query performance prediction for neural ir: Are we there yet? In Jaap Kamps, Lorraine Goeuriot, Fabio Crestani, Maria Maistro, Hideo Joho, Brian Davis, Cathal Gurrin, Udo Kruschwitz, and Annalina Caputo, editors, *Advances in Information Retrieval - 45th European Conference on Information Retrieval, ECIR 2023, Dublin, Ireland, April 2-6, 2023, Proceedings, Part I*, volume 13980 of *Lecture Notes in Computer Science*, pages 232–248. Springer, 2023.

[4] Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. Precise zero-shot dense retrieval without relevance labels. *arXiv preprint arXiv:2212.10496*, 2023.

[5] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *ICML*, 2017.

[6] Mohanna Hoveyda, Jelle Piepenbrock, Arjen P. de Vries, Maarten de Rijke, and Faegheh Hasibi. Orlog: Resolving complex queries with llms and probabilistic reasoning. In Ricardo Campos, Adam Jatowt, Yanyan Lan, Mohammad Aliannejadi, Christine Bauer, Sean MacAvaney, Avishek Anand, Zhaochun Ren, Suzan Verberne, Nan Bai, and Masoud Mansoury, editors, *Advances in Information Retrieval - 48th European Conference on Information Retrieval, ECIR 2026, Delft, The Netherlands, March 29 - April 2, 2026, Proceedings, Part I*, volume 16483 of *Lecture Notes in Computer Science*, pages 98–114. Springer, 2026.

[7] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *EMNLP*, 2020.

[8] Taegyeong Lee, Jiwon Park, Seunghyun Hwang, and JooYoung Jang. Deo: Training-free direct embedding optimization for negation-aware retrieval. *arXiv preprint arXiv:2603.09185*, 2026.

[9] Xiang Li et al. Deepretrieval: Retrieval-augmented query generation for information retrieval. *arXiv preprint*, 2025.

[10] Xinbei Ma, Yeyun Gong, Pengcheng He, Hai Zhao, and Nan Duan. Query rewriting for retrieval-augmented large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023.

[11] Niklas Muennighoff et al. Generative representational instruction tuning. *arXiv preprint arXiv:2402.09906*, 2024.

[12] Niklas Muennighoff, Nouamane Tazi, Loec Magne, and Nils Reimers. Mteb: Massive text embedding benchmark. In *EACL*, 2023.

[13] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *NeurIPS*, 2022.

[14] John Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers*, 1999.

[15] Aashish Pokhrel and Shivanand Venkanna Sheshappanavar. Negation matters: Training-free negation-aware image retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 11353–11362, jun 2026.

[16] Simon Raudaschl. Rag-fusion: A new take on retrieval-augmented generation, 2024.

[17] Victor Sanh, Albert Webson, Colin Raffel, Stephen Bach, Linting Patry, et al. Multitask prompted training enables zero-shot task generalization. *ICLR*, 2022.

[18] Hong Su et al. One embedder, any task: Instruction-finetuned text embeddings. In *NeurIPS*, 2022.

[19] Nandan Thakur, Nils Reimers, Andreas Rücklé, Iryna Gurevych, and Johannes Daxenberger. Beir: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *NeurIPS Datasets and Benchmarks*, 2021.

[20] Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, et al. Improving text embeddings with large language models. *arXiv preprint arXiv:2401.00368*, 2023.

[21] Xiao Wang, Craig Macdonald, Nicola Tonellotto, and Iadh Ounis. Pseudo-relevance feedback for multiple representation dense retrieval. In Faegheh Hasibi, Yi Fang, and Akiko Aizawa, editors, *ICTIR '21: The 2021 ACM SIGIR International Conference on the Theory of Information Retrieval*, pages 297–306. ACM, 2021.

[22] Zhicheng Wang, Yujia Zhang, Cheng Lu, Xiang Zhang, et al. Active retrieval augmented generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023.

[23] Jason Wei, Maarten Bosma, Vincent Y. Zhao, Kelvin Guu, Adams Yu, Brian Dunselman, Quoc V. Dai, and Ed H. Le. Finetuned language models are zero-shot learners. *ICLR*, 2022.

[24] Orion Weller, Benjamin Chang, Sean MacAvaney, Kyle Lo, Arman Cohan, Benjamin Van Durme, Dawn Lawrie, and Luca Soldaini. Followir: Evaluating and teaching information retrieval models to follow instructions. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 11926–11942, Albuquerque, New Mexico, apr 2025. Association for Computational Linguistics.

[25] Orion Weller, Benjamin Van Durme, Dawn Lawrie, Ashwin Paranjape, Yuhao Zhang, and Jack Hessel. Promptriever: Instruction-trained retrievers can be prompted like language models. In *International Conference on Learning Representations*, 2025.

[26] Ganlin Xu, Zhoujia Zhang, Wangyi Mei, Jiaqing Liang, Weijia Lu, Xiaodong Zhang, Zhifei Yang, Xiaofeng Ma, Yanghua Xiao, and Deqing Yang. Logical consistency is vital: Neural-symbolic information retrieval for negative-constraint queries. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Findings of the Association for Computational Linguistics: ACL 2025*, pages 1828–1847, Vienna, Austria, jul 2025. Association for Computational Linguistics.

[27] Qingcheng Zeng et al. Dual-view training for instruction-following information retrieval. *arXiv preprint arXiv:2604.18845*, 2026.

[28] Han Zhou, Xingchen Wan, Lev Proleev, Diana Mincu, Jilin Chen, Katherine A. Heller, and Subhrajit Roy. Batch calibration: Rethinking calibration for in-context learning and prompt engineering. In *The Twelfth International Conference on Learning Representations*. OpenReview.net, 2024.

[29] Yuchen Zhuang, Aaron Trinh, Rushi Qiang, Haotian Sun, Chao Zhang, Hanjun Dai, and Bo Dai. Towards better instruction following retrieval models. *arXiv preprint arXiv:2505.21439*, 2025.